Glass Temperature Controller Plan

Sep 28, 2026 · @Lauren

1. The plan in one page

The glass gets its own PI controller, designed exactly like the chamber one: lambda tuning with $\lambda = 60$ s ($K_p = 0.0361$, $K_i = 2.28 \times 10^{-4}$) and setpoint weighting $b = 0.5$, holding the glass at 39 °C.

The glass is about four times faster than the chamber ($\tau = 158$ s vs 594 s), so its loop is tuned faster too. That way it reaches 39 °C in under 10 min, well before the chamber reaches 37 °C, and it stays at least 2 °C above the chamber the whole time, which is what stops condensation.

The order to work in:

- [ ] Tidy up the glass model wording (section 2).
- [ ] Design the PI with lambda tuning, $\lambda = 60$ s (section 3).
- [ ] Add setpoint weighting $b = 0.5$ (section 4).
- [ ] Check the interaction with the chamber loop (section 5).
- [ ] Check stability and delay (section 6).
- [ ] Add the glass loop to your Simulink model (section 7).
- [ ] Run the tests and save the plots (section 8).
- [ ] Once the glass is built: measure its resistance, do a step test and retune (section 9).

Everything here follows the same method as the chamber plan, so if a step feels unfamiliar, the chamber doc explains it in more detail.

2. The glass model

Your glass model is correct: the numbers check out, and the three input gains add up to 1 ($0.383 + 0.616 \approx 1$), which is the same consistency check the chamber model passed.

The heat balance

```
C_g \frac{dT_g}{dt} = \mu_g P_g + \frac{T_c - T_g}{R_{cg}} - \frac{T_g - T_a}{R_{ga}}
```

Symbol	Meaning	Value	Check
C_g	Glass capacitance, 150 × 112 × 1.1 mm, ρ = 2500, c_p = 750	34.7 J/K	✓
P_g	ITO heater power at full duty (nominal)	16 W	✓ (range 10.7 to 21.5 W)
R_cg	Glass to chamber (same path as in the chamber model)	11.9 K/W	✓
R_ga	Glass to room, 1/(h_a A_glass)	7.4 K/W	✓
1/R_cg + 1/R_ga	Total loss conductance	0.219 W/K	✓

The transfer function

$$T_g(s) = \frac{73\,\mu_g(s) + 0.383\,T_c(s) + 0.616\,T_a(s)}{158.3s + 1}$$

So the plant the glass controller sees is:

$$G_g(s) = \frac{K_g}{\tau_g s + 1} = \frac{73}{158.3s + 1}$$

- **K_g = 73 K:** full ITO power would hold the glass 73 °C above its rest point. Holding 39 °C needs about **15% duty** (about 23% if the glass turns out to be the 10.7 W end of the range).
- **τ_g = 158 s:** about 2.6 min, roughly four times faster than the chamber. Glass is thin and light, so this makes sense.
- **The chamber is a disturbance here** (gain 0.383), just as the glass was a disturbance for the chamber.

Small changes worth making to the write-up

- **"It has two heat paths only"** reads oddly when the next sentence adds a third. Say "Two heat paths are modelled" instead.
- **R_ga may be on the high side.** Glass is a good radiator (emissivity about 0.9), and radiation alone from its top face is worth roughly 5 to 6 W/m²K. So the combined coefficient could be nearer 10 to 12 W/m²K than 8. That would mean more power to hold 39 °C and a slightly shorter τ. It doesn't change the design, and the step test will show it, but one sentence acknowledging it is good.

- **R_cg uses the full glass area,** but the edges rest on the wall, so the area facing the chamber is a bit smaller. This is minor and consistent with the chamber model, so leave it.
- **Say the Laplace variables are deviations** around the operating point, the same as the chamber model.

3. The controller: PI with lambda tuning

The glass controller is a PI with Kp = 0.0361 and Ki = 2.28×10^{-4}, which makes the glass loop a first-order system with time constant λ = 60 s.

PI (no D) for the same reasons as the chamber: the plant is first order, temperature is slow, and D would only amplify sensor noise.

Step 1: choose the response you want

We want the glass to rise smoothly to 39 °C, so the target closed loop is first order:

```
T(s) = \frac{1}{\lambda s + 1}
```

Choosing λ = 60 s. Three things set this:

- **Faster than the chamber loop (λ = 300 s).** The glass should reach temperature first, so the chamber sees it as a steady input and the glass stays above the chamber. Five times faster is a comfortable gap.
- **Not too aggressive.** λ = 60 s is about τ_g/2.6. The warm-up duty starts at about 0.31, so the heater never saturates.
- **Room for delay.** A sensor on glass responds in a few seconds, and the loop tolerates up to about 94 s (section 6).

Step 2: find the loop that gives that response

For unity feedback, T = L/(1 + L). Solving for L gives L = T/(1 − T), which for a first-order T is:

```
L(s) = \frac{1}{\lambda s} = \frac{1}{60s}
```

Step 3: solve for the controller

L = C × G, so C = L ÷ G:

```
C_g(s) = \frac{1}{60s} \div \frac{73}{158.3s + 1} = \frac{1}
{60s}\cdot\frac{158.3s + 1}{73} = \frac{158.3s + 1}{4380\,s}
```

(60 × 73 = 4380.)

Step 4: split it into P and I

A PI controller is Kp + Ki/s. Split the fraction into two pieces:

$$C_g(s) = \frac{158.3s}{4380s} + \frac{1}{4380s} = 0.0361 + \frac{2.28\times10^{-4}}{s}$$

Read off the gains. The general formulas are the same as for the chamber:

Parameter	Formula	Value
Kp	$\tau_g / (K_g \lambda) = 158.3 / (73 \times 60)$	0.0361 (3.6% duty per °C)
Ki	$1 / (K_g \lambda) = 1 / (73 \times 60)$	2.28×10^{-4}
Ti	Kp / Ki = τ_g	158.3 s

Step 5: check the loop and closed loop

Multiply the controller by the plant. The (158.3s + 1) terms cancel:

$$L(s) = \frac{158.3s + 1}{4380s}\cdot\frac{73}{158.3s + 1} = \frac{73}{4380s} = \frac{1}{60s}$$

$$\frac{T_g(s)}{T_{g,sp}(s)} = \frac{L}{1 + L} = \frac{1}{60s + 1}$$

First order, time constant 60 s, gain 1: no overshoot on setpoint changes (in the linear model), zero steady-state error, and essentially at setpoint after 4λ = 4 min.

Faster option: if the step test shows almost no delay, λ = 40 s also works well (Kp = 0.0542, Ki = 3.42×10^{-4}). In simulation it settles in about 9 min with 0.08 °C glass overshoot, and the warm-up duty starts at 0.46.

4. Overshoot and setpoint weighting

Setpoint weighting b = 0.5 cuts the glass overshoot from 0.39 °C to 0.14 °C and settles it faster, and a small glass overshoot is harmless.

Why the glass overshoots even with b = 0.5

For the chamber, b = 0.5 removed the overshoot completely. For the glass it doesn't, and the reason is the chamber. The glass reaches 39 °C in under 10 min, but the chamber keeps warming for another 30 min after that. A warmer chamber means less heat leaks

out of the glass, so the chamber slowly pushes the glass up, and the glass loop has to keep backing off. That shows up as a small, slow overshoot.

Why that's fine

The glass has a looser job than the chamber. It only needs to stay above the chamber to stop condensation. A glass at 39.1 °C does no harm to the samples, and it very slightly helps the chamber warm up. The overshoot you must avoid is in the chamber, and that stays at zero in every case.

Results with both loops running

Warm-up from 22 °C, both loops starting together, λ_g = 60 s:

Setpoint weight b	Glass overshoot	Glass settles (±0.2 °C)	Glass duty at start	Chamber overshoot	Smallest glass-to-chamber gap
1 (normal PI)	0.39 °C	14 min	0.61	0.00 °C	2.0 °C
0.5 (chosen)	0.14 °C	8 min	0.31	0.00 °C	2.0 °C
0 (P on measurement only)	0.11 °C	11 min	0.00	0.00 °C	**−0.07 °C**

b = 0 is a trap: the glass starts with zero duty and falls **behind** the chamber for a while, which is exactly when condensation would form. So b = 0.5 is the right balance.

Setpoint transfer function

The same form as the chamber:

$$\frac{T_g(s)}{T_{g,sp}(s)} = \frac{b\,\tau_g s + 1}{(\tau_g s + 1)(\lambda s + 1)} = \frac{79.2s + 1}{(158.3s + 1)(60s + 1)}$$

As before, use temperatures relative to ambient in the P path: the weighted setpoint is $b \cdot T_{g,sp} + (1 - b) \cdot T_a = 0.5 \times 39 + 0.5 \times 22 = 30.5$ °C, and the full measured T_g is subtracted from it.

5. Working with the chamber loop

The two loops can be designed separately and run together: they interact only weakly (relative gain 1.13), the chamber still never overshoots, and the glass stays at least 1.8 °C above the chamber in every case tested.

Each loop sees the other as a disturbance

- The chamber loop sees the glass through the term 0.299·T_g.
- The glass loop sees the chamber through the term 0.383·T_c.

Each controller's integrator removes a steady disturbance completely, so once both are settled, neither affects the other's final temperature.

Checking the interaction: the relative gain array

The relative gain array (RGA) is the standard check for whether two loops can be designed on their own. It compares the gain from each heater to its own temperature with the other loop open and closed. A value of 1 means no interaction.

From the steady-state heat balances of both nodes:

```
\lambda_{11} = \frac{g_{11}g_{22}}{g_{11}g_{22} - g_{12}g_{21}} = 1.13
```

That is close to 1: each heater mostly affects its own temperature, so pairing film heaters with the chamber and the ITO with the glass is right, and separate design is justified. Put this number in the report.

The warm-up order looks after itself

Because the glass is faster (τ = 158 s) and tuned faster (λ = 60 s), it reaches 39 °C in about 8 min while the chamber takes about 38 min. So you can start both loops at the same time. The glass leads throughout, which is what prevents condensation.

Robustness with both loops running

λ_g = 60 s, b = 0.5 on both loops, warm-up from 22 °C:

Case	Glass overshoot	Glass settles	Chamber overshoot	Smallest gap
Nominal (P_g = 16 W, C_g = 34.7 J/K)	0.14 °C	8 min	0.00 °C	2.0 °C
Weak ITO (P_g = 10.7 W)	0.24 °C	17 min	0.00 °C	1.9 °C
Strong ITO (P_g = 21.5 W)	0.09 °C	9 min	0.00 °C	2.0 °C
Lighter glass (C_g = 25 J/K)	0.10 °C	9 min	0.00 °C	2.0 °C
Heavier glass (C_g = 50 J/K)	0.29 °C	15 min	0.00 °C	2.0 °C

Case	Glass overshoot	Glass settles	Chamber overshoot	Smallest gap
Ambient drops 4 °C after warm-up	recovers	recovers	0.00 °C	1.8 °C

The ITO power (10.7 to 21.5 W depending on sheet resistance) is the biggest unknown, and the design copes with the whole range. At 10.7 W the glass needs about 23% duty to hold 39 °C, still well within range.

One firmware rule worth adding

If the glass heater or its sensor fails, the glass could drop below the chamber and fog. A simple check that raises a warning when $T_g < T_c + 0.5$ °C is cheap insurance.

6. Stability, sampling and delay

The glass loop is stable with a 90° phase margin and tolerates about 94 s of delay; sampling at 1 s costs under 2° of that.

Check	Result	What it means
Closed-loop poles	$s = -1/60$ and $s = -1/158.3$	Real and negative, so stable with no oscillation
Phase margin	90°	$L(s) = 1/(60s)$ has −90° phase everywhere
Crossover frequency	1/60 = 0.0167 rad/s	Five times higher than the chamber loop's 0.0033 rad/s
Delay margin	about 94 s ($= \lambda \cdot \pi/2$)	How much lag it takes to go unstable
Sampling at 1 s	about 1.5 s delay, costs 1.4°	Phase margin 88.6°; rule of thumb $T_s < \lambda/10 = 6$ s is met

Delay matters more here than for the chamber

The glass loop is five times faster, so the same delay eats five times more phase. It is still comfortable, but sensor mounting matters:

Sensor lag	Phase margin
1.5 s (sampling only)	88.6°

Sensor lag	Phase margin
10 s (thermistor glued to glass)	80°
30 s (sensor in a housing, poor contact)	61°

Anything above about 45° is healthy. So: bond the glass sensor directly to the glass with thermal paste or thermally conductive adhesive, and keep it small. Put it near the centre of the viewing area, not at an edge, because the edges run cooler where the glass rests on the wall.

Practical points

- **PWM resolution:** the glass holds at about 15% duty, so 10-bit PWM or better, as for the chamber.
- **Over-temperature cutoff:** turn the ITO off above about 45 °C. ITO heats fast (at full power the glass starts rising at about 25 °C per minute), so a stuck-on heater needs catching quickly.
- **Integral stored as a float,** same as the chamber (Ki × error × 1 s is only about 2×10^{-4} per °C).

Discrete-time check (z-domain)

The glass controller also runs digitally at Ts = 1 s, and the z-domain check confirms the digital loop is stable with poles matching the continuous design.

Design by emulation is valid here because Ts < λ/10 = 6 s. The margin is smaller than for the chamber (6 s vs 30 s) because the glass loop is faster, but Ts = 1 s still meets it comfortably.

Plant with zero-order hold:

```
G_g(z) = \frac{K_g(1 - a)}{z - a}, \qquad a = e^{-T_s/\tau_g} = e^{-1/158.3}
= 0.993703 \quad\Rightarrow\quad G_g(z) = \frac{0.4597}{z - 0.993703}
```

Controller exactly as coded:

```
C_g(z) = K_p + \frac{K_i T_s}{z - 1} = 0.0361 + \frac{2.28\times10^{-4}}{z -
1}
```

Closed-loop poles:

Discrete pole	Continuous pole mapped to z = e^(sTs)	Match
0.983418	e^(−1/60) = 0.983471	✓
0.993671	e^(−1/158.3) = 0.993703	✓

Both are inside the unit circle, so the digital glass loop is stable, and they match the continuous design closely. The match is slightly less exact than for the chamber because the glass loop is faster relative to the 1 s sample, which is expected.

In MATLAB:

```
Kg = 73; taug = 158.3; lamg = 60;
Kpg = taug/(Kg*lamg); Kig = 1/(Kg*lamg);
s = tf('s'); Ts = 1;
Gg  = Kg/(taug*s + 1);
Tg_c = feedback((Kpg + Kig/s)*Gg, 1);    % continuous
Ggz = c2d(Gg, Ts, 'zoh');
z   = tf('z', Ts);
Cgz = Kpg + Kig*Ts/(z - 1);
Tgz = feedback(Cgz*Ggz, 1);              % discrete
disp(abs(pole(Tgz)))                     % all < 1 -> stable
figure; step(17*Tgz, 17*Tg_c, 600);
legend('Discrete (Ts = 1 s)', 'Continuous'); grid on;
```

7. Building it in Simulink

The glass loop is a copy of your chamber loop with different numbers, and the only new wiring is that the glass output replaces the constant 39 in the chamber, and the chamber output feeds into the glass.

Step 1: the glass plant

Build it the same way as your chamber, from the differential equation:

```
\frac{dT_g}{dt} = \frac{73\,u_g + 0.383\,T_c + 0.616\,T_a - T_g}{158.3}
```

1. u_g (from the glass Saturation block) → **Gain** `73`
2. T_c (from your chamber Integrator output) → **Gain** `0.383`
3. **Constant** T_a (use the same value as the chamber, e.g. 20 or 22) → **Gain** `0.616`
4. A **Sum** with signs `+++-` : the three gains, plus T_g fed back into the minus input
5. → **Gain** `1/158.3`

6. → **Integrator**, Initial condition = T_a (the glass starts at room temperature)
7. The Integrator output is **T_g**

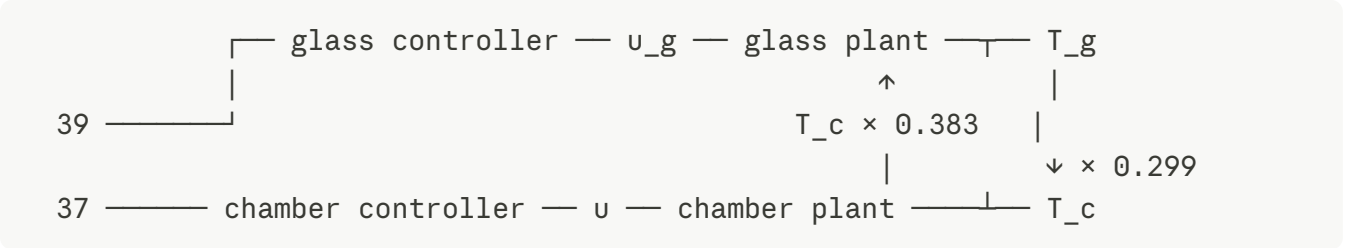
Step 2: the glass controller

Exactly the same layout as your chamber controller:

1. **I path:** Constant `39` and T_g into a **Sum** (`+-`) → **Gain** `2.28e-4` → **Integrator** (initial condition 0, Limit output ticked, 0 to 1)
2. **P path:** Constant `0.5*39 + (1-0.5)*Ta` (with your Ta value typed in) and T_g into a **Sum** (`+-`) → **Gain** `0.0361`
3. Add the P and I paths in a **Sum** (`++`) → **Saturation** (0 to 1) → this is u_g

Step 3: connect the two loops

1. In the chamber, **delete the Constant** `39` that feeds the `0.299` gain, and connect **T_g** there instead.
2. Connect **T_c** into the glass `0.383` gain (step 1.2 above).



The two plants are now coupled both ways, exactly like the real hardware.

Step 4: outputs and settings

- Put T_c and T_g on the same **Scope** (a Mux, or a scope with 2 inputs) so you can see the glass lead the chamber.
- A second scope on u and u_g.
- A third signal worth plotting: **T_g − T_c** (one Sum block). This is your condensation margin, and it should stay positive.
- Stop time `4000` s as before.

Values to expect

Signal	Start	Settles at
T_g	Ta	39 °C in about 8 min, peak about 39.14 °C
T_c	Ta	37 °C in about 38 min, no overshoot

Signal	Start	Settles at
u_g	about 0.31	about 0.15 (Ta = 22)
u	about 0.12	about 0.077 (Ta = 22)
T_g − T_c	0	2.0 °C, never negative

With Ta = 20 instead of 22, both duties settle slightly higher.

8. Tests to run

Five simulations show the glass loop works on its own terms and does not upset the chamber.

Test	What to change	What it proves	What to expect
1. Coupled warm-up	Nothing, both loops start together from Ta	Glass leads, chamber still has no overshoot	T_g at 39 in about 8 min, T_c at 37 in about 38 min, T_g − T_c never negative
2. Glass fixed vs glass controlled	Rerun with the Constant 39 back in place of T_g	Treating the glass as a disturbance was justified	Chamber curves almost on top of each other
3. Setpoint weight	Glass b = 1, 0.5, 0	Why b = 0.5	0.39, 0.14 and 0.11 °C glass overshoot; b = 0 lets the glass fall behind the chamber
4. Ambient disturbance	Ta steps 22 → 18 after warm-up	Both loops recover	T_g − T_c dips to about 1.8 °C, then recovers
5. Robustness	P_g = 73 × 10.7/16 and 73 × 21.5/16 in the plant gain (49 and 98)	Copes with the unknown ITO power	Glass settles in 9 to 17 min, chamber unaffected

For test 5, only change the `73` gain in the **plant**, not the controller gains. That is what happens in reality: you tuned for 16 W, but the glass turns out different.

The plot that tells the whole story for the report is test 1 with T_c, T_g and T_g − T_c on one figure.

9. Once the glass is built

Measure the ITO resistance first, then do one step test; together they give the real K_g, τ_g and delay, which go straight into the same formulas.

Step 1: measure the resistance (5 minutes)

With the copper tape on, measure across the two electrodes with a multimeter. Then:

$$P_g = \frac{V^2}{R_{glass}} \qquad K_g \approx P_g \times 4.56\ \text{K/W}$$

(4.56 K/W is 1/0.219, the glass's total loss resistance.) This alone tells you whether you're at the 10.7 W or 21.5 W end, and you can update the Simulink plant gain immediately.

Step 2: the step test

1. Hold the chamber steady, ideally with the chamber loop running and settled at 37 °C. That way T_c is constant and the test measures only the glass.
2. Let the glass sit with the ITO off until its temperature is steady.
3. Switch the ITO to a **fixed** duty of about 20% (no controller) and log T_g every second.
4. Log until it stops rising, about 15 min (5τ_g is about 13 min).

Step 3: read off the numbers

- **K_g** = temperature rise / duty step (e.g. a 14.6 °C rise at 20% gives K_g = 73)
- **θ** = time before the temperature starts moving, likely only a few seconds
- **τ_g** = time to reach 63% of the rise, measured from the end of the delay

Step 4: retune

$$K_p = \frac{\tau_g}{K_g(\lambda + \theta)} \qquad K_i = \frac{1}{K_g(\lambda + \theta)}$$

Keep λ = 60 s unless the delay is large. Keep λ at least 2θ, and keep it at least 3 to 5 times faster than the chamber's λ so the glass keeps leading. If the delay is tiny, λ = 40 s is a good faster option.

What changes and what doesn't

Stays the same	Gets updated
PI structure, b = 0.5, clamp and anti-windup	K_g, τ_g from the measurements

Stays the same	Gets updated
Simulink layout and the coupling to the chamber	Kp and Ki for the glass
Stability argument, RGA check	Possibly λ_g, if there is a noticeable delay

For the report: glass model, lambda tuning, coupled simulation showing the glass leads the chamber, RGA, step test, retuned gains.